Aishwarya (Aishu) Kamal

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SUMMARY

Distributed-systems leader with a decade at Amazon and Google building planet-scale database and AI infrastructure, specializing in **distributed consensus, replication, and failover** — from Paxos modifications that took DynamoDB to five nines, to the quorum architectures behind Memorystore's cross-region replication and single-shard offerings, to resilient accelerator multiplexing for RL. I lead the reinforcement-learning infrastructure optimization space for Google Kubernetes Engine and drive RL-infrastructure direction in the open ecosystem (llm-d, SIG-RL); previously led marquee launches for Amazon DynamoDB and Google Cloud Memorystore. Named inventor on two granted U.S. patents cited as prior art in subsequent Oracle, IBM, Snowflake, and Harvard patent prosecutions.

EXPERIENCE

Staff Software Engineer — RL Infrastructure Lead, Google Kubernetes Engine
Nov 2025 – present

Google

- Leading the **reinforcement-learning infrastructure optimization space** for GKE AI workloads — setting direction across accelerator utilization, autoscaling, and resilience for large-scale RL post-training and LLM inference; driving RL-infrastructure direction in llm-d's **SIG-RL** community.
- Architected the industry's first **co-operative accelerator time-slicing platform** for RL — designed to drive accelerator utilization above 95% — in **llm-d** (CNCF sandbox; founded by Google, IBM, Red Hat, NVIDIA, CoreWeave; shipped in Red Hat OpenShift AI 3.0 and on AWS) — see the Google Cloud announcement and the platform on GitHub, which I lead and maintain.
- Architected llm-d's **workload autoscaling** well-lit path — scale-from-zero and HPA-based autoscaling, the project's core autoscaling story — now maintained with IBM, Red Hat, Intel, and Solo.io engineers.

Technical Lead — Google Cloud Memorystore
Aug 2021 – Nov 2025

Google

- Led design and launch of **Memorystore for Redis Cluster** (unveiled at Google Cloud Next '23; GA with 99.99% SLA) — a new product line for Google Cloud — adopted in production by Reddit, Character.AI, Unity Ads (10M ops/sec), Statsig (7.5M peak QPS), Palo Alto Networks, and NTT DOCOMO.
- Led launch of **Memorystore for Valkey** — the first major-cloud fully managed Valkey 8.0 service following the 2024 Redis license fork — adopted by MLB, Snap, Mercado Libre, and Juspay (GPay/UPI payment stack for leading Indian banks).
- Continued my consensus-and-replication specialization: led **cross-region replication** for Redis Cluster and Valkey — cited publicly by customers (MLB; Buildertrend, which named the feature as its migration driver across ~300 instances) — and architected the standalone **single-shard offering** (2025) on a voting-replica and auxiliary-node quorum design that closed the failover-availability gap in open-source Redis/Valkey, opening a second product tier.

Software Engineer → Senior Software Engineer (Tech Lead) — Amazon DynamoDB
Feb 2017 – Aug 2021

Amazon Web Services

- Led the design and launch of **Transactions** (re:Invent '18 — brought ACID to serverless NoSQL for one of the largest database services on earth at 126M peak requests/sec; a shift competitors followed and the subject of the USENIX ATC '23 Best Paper), and of the **Standard-IA table class** (announced re:Invent '21 — cut storage costs up to 60%).
- Architected and patented the data model behind **Amazon Keyspaces**, carrying it from inception (re:Invent '19) to GA (2020) — an entirely new AWS product line, trusted today as the primary store for Monzo Bank's core banking ledger (350 TB, 2,000+ tables) and adopted by Intuit, Adobe, and GE Vernova.
- Led the availability program that improved DynamoDB availability by an order of magnitude, from 99.99% to 99.999% — a 10× reduction in permitted downtime — through deep modifications to the Paxos consensus layer.
- Earlier, during DynamoDB's hyper-growth years, shipped its re:Invent-headline data-protection wave as a core engineer: **Time-to-Live** (2017), petabyte-scale **Backup & Restore** (re:Invent '17), and **Point-in-Time Recovery** (2018) — together, the "undo button" for cloud data.

PATENTS & SELECTED WRITING

- **US 11,036,762** — Compound partition and clustering keys (2021): the data model underlying Amazon Keyspaces. Cited as prior art in subsequent prosecutions by Harvard, Oracle, and IBM.
- **US 11,138,164** — Alter table implementation details with schema versioning (2021): zero-downtime online schema evolution. Cited as prior art in prosecutions by Snowflake, Oracle, Dell, and FIS.
- **"Introducing co-operative time-slicing for RL in llm-d"** — Google Cloud Blog (2026), co-author; independently covered by trade press.
- **"RL Post-Training: Co-Operative Time-Slicing with llm-d"** — llm-d project blog (2026), co-author; accepted and published by project maintainers at IBM Research and Red Hat.

COMMUNITY LEADERSHIP & PROFESSIONAL SERVICE

- Maintainer and code owner, llm-d-rl-time-slicing; reviewer and approver across llm-d projects.
- A leading technical voice in llm-d's SIG-RL (reinforcement-learning workloads special interest group), helping shape its agenda and direction.
- Graduate Teaching Assistant, UCLA Computer Science (2015–2016).

EDUCATION

- **M.S., Computer Science** — University of California, Los Angeles (UCLA), 2015–2016.
- **B.E., Computer Science** — Visvesvaraya Technological University, India, 2011–2015.